

```
In [1]: import pandas as pd
df = pd.read_excel("D:\\Data Engineering\\Python\\Bike_Data.xlsx")
```

```
In [2]: df.head()
```

Out[2]:

	Region	Country	Customer	Business Segment	Category	Model	Color	SalesDate	ListF
0	North America	United States	Advanced Bike Components	Components	Road Frames	LL Road Frame	Red	2020-04-01	33
1	North America	United States	Central Discount Store	Bikes	Mountain Bikes	Mountain-100	Silver	2020-04-01	339
2	North America	United States	Leading Sales & Repair	Clothing	Jerseys	Long-Sleeve Logo Jersey	Multi	2020-04-01	4
3	North America	United States	Paint Supply	Components	Mountain Frames	HL Mountain Frame	Black	2020-04-01	134
4	North America	United States	Scooters and Bikes Store	Bikes	Road Bikes	Road-450	Red	2020-04-01	145

```
In [4]: import pandas as pd
data = pd.read_csv("D:\\Data Engineering\\Python\\car_data.csv")
```

```
In [6]: data.shape # to get size of dataset
```

Out[6]: (301, 8)

```
In [7]: print("The dataset has :",data.shape[0], "rows")
```

The dataset has : 301 rows

```
In [8]: print("The dataset has :",data.shape[1],"columns")
```

The dataset has : 8 columns

```
In [9]: data.head()
```

```
Out[9]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tran
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	



```
In [10]: data.tail()
```

```
Out[10]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tr
296	city	2016	9.50	11.6	33988	Diesel	Dealer	
297	brio	2015	4.00	5.9	60000	Petrol	Dealer	
298	city	2009	3.35	11.0	87934	Petrol	Dealer	
299	city	2017	11.50	12.5	9000	Diesel	Dealer	
300	brio	2016	5.30	5.9	5464	Petrol	Dealer	



```
In [11]: data.columns #It'll give you all columns present in dataset
```

```
Out[11]: Index(['Car_Name', 'Year', 'Selling_Price', 'Present_Price', 'Kms_Drive  
n',  
               'Fuel_Type', 'Seller_Type', 'Transmission'],  
              dtype='object')
```

```
In [12]: list(data.columns)
```

```
Out[12]: ['Car_Name',  
          'Year',  
          'Selling_Price',  
          'Present_Price',  
          'Kms_Driven',  
          'Fuel_Type',  
          'Seller_Type',  
          'Transmission']
```

```
In [13]: data.dtypes #It'll give you datatypes of all columns
```

```
Out[13]: Car_Name      object
Year          int64
Selling_Price  float64
Present_Price  float64
Kms_Driven     int64
Fuel_Type      object
Seller_Type    object
Transmission   object
dtype: object
```

```
In [15]: data[["Car_Name", "Year", "Selling_Price"]].head()
```

```
Out[15]:
```

	Car_Name	Year	Selling_Price
0	ritz	2014	3.35
1	sx4	2013	4.75
2	ciaz	2017	7.25
3	wagon r	2011	2.85
4	swift	2014	4.60

```
In [17]: data[data["Year"] == 2014].head() #1 approach to filter the records
```

```
Out[17]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tra
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	
23	alto k10	2014	2.50	3.46	45280	Petrol	Dealer	
32	swift	2014	4.95	7.49	39000	Diesel	Dealer	
33	ertiga	2014	6.00	9.95	45000	Diesel	Dealer	

```
In [18]: data[data.Year == 2014].head() #2nd approach to filter the records
```

```
Out[18]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Tra
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	
23	alto k10	2014	2.50	3.46	45280	Petrol	Dealer	
32	swift	2014	4.95	7.49	39000	Diesel	Dealer	
33	ertiga	2014	6.00	9.95	45000	Diesel	Dealer	

```
In [19]: data[data["Transmission"] == "Manual"].head() #with one condition
```

```
Out[19]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Trar
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	

```
In [21]: data[(data["Year"] == 2014) & (data["Transmission"] == "Manual")].head() #
```

```
Out[21]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Trar
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	
23	alto k10	2014	2.50	3.46	45280	Petrol	Dealer	
32	swift	2014	4.95	7.49	39000	Diesel	Dealer	
33	ertiga	2014	6.00	9.95	45000	Diesel	Dealer	

```
In [22]: data[(data["Year"] == 2014) & (data["Transmission"] == "Manual") & (data["
```

```
Out[22]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Trar
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	
23	alto k10	2014	2.50	3.46	45280	Petrol	Dealer	
41	alto k10	2014	2.55	3.98	46706	Petrol	Dealer	
58	etios g	2014	4.10	6.80	39485	Petrol	Dealer	
89	etios g	2014	4.75	6.76	40000	Petrol	Dealer	

```
In [23]: data[(data["Year"] == 2014) & (data["Kms_Driven"] <= 15000)].head(5) # to
```

```
Out[23]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Ti
45	ciaz	2014	7.50	12.040	15000	Petrol	Dealer	
112	KTM 390 Duke	2014	1.15	2.400	7000	Petrol	Individual	
146	Hero Extreme	2014	0.55	0.787	15000	Petrol	Individual	
154	Yamaha Fazer	2014	0.50	0.880	8000	Petrol	Individual	
172	Bajaj Pulsar 135 LS	2014	0.40	0.640	13700	Petrol	Individual	

```
In [ ]: #to get 2014 and 2015 cars along with automatic trasmission
```

```
In [27]: data[(data["Year"].isin([2014,2015])) & (data["Transmission"] == "Automatic")]
```

```
Out[27]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Ti
12	ciaz	2015	7.50	9.94	15000	Petrol	Dealer	
45	ciaz	2014	7.50	12.04	15000	Petrol	Dealer	
51	fortuner	2015	23.00	30.61	40000	Diesel	Dealer	
59	fortuner	2014	19.99	35.96	41000	Diesel	Dealer	
62	fortuner	2014	18.75	35.96	78000	Diesel	Dealer	
63	fortuner	2015	23.50	35.96	47000	Diesel	Dealer	
93	fortuner	2015	23.00	30.61	40000	Diesel	Dealer	
178	TVS Jupyter	2014	0.35	0.52	19000	Petrol	Individual	
207	grand i10	2015	5.15	5.70	34797	Petrol	Dealer	
232	elantra	2015	11.45	14.79	12900	Petrol	Dealer	

```
In [ ]: #to get the multiple filter conditions and with selected columns
```

```
In [28]: data[(data["Year"].isin([2014,2015])) & (data["Transmission"] == "Automati
```

```
Out[28]:
```

	Car_Name	Year	Selling_Price
12	ciaz	2015	7.50
45	ciaz	2014	7.50
51	fortuner	2015	23.00
59	fortuner	2014	19.99
62	fortuner	2014	18.75

```
In [29]: list(data.columns)
```

```
Out[29]: ['Car_Name',  
          'Year',  
          'Selling_Price',  
          'Present_Price',  
          'Kms_Driven',  
          'Fuel_Type',  
          'Seller_Type',  
          'Transmission']
```

```
In [33]: data[(data["Year"].isin([2015,2017,2018])) & (data["Fuel_Type"] == "Petrol
```

```
Out[33]:
```

	Car_Name	Year	Selling_Price
2	ciaz	2017	7.25
6	ciaz	2015	6.75
10	alto 800	2017	2.85
12	ciaz	2015	7.50
13	ertiga	2015	6.10